

3D stuff

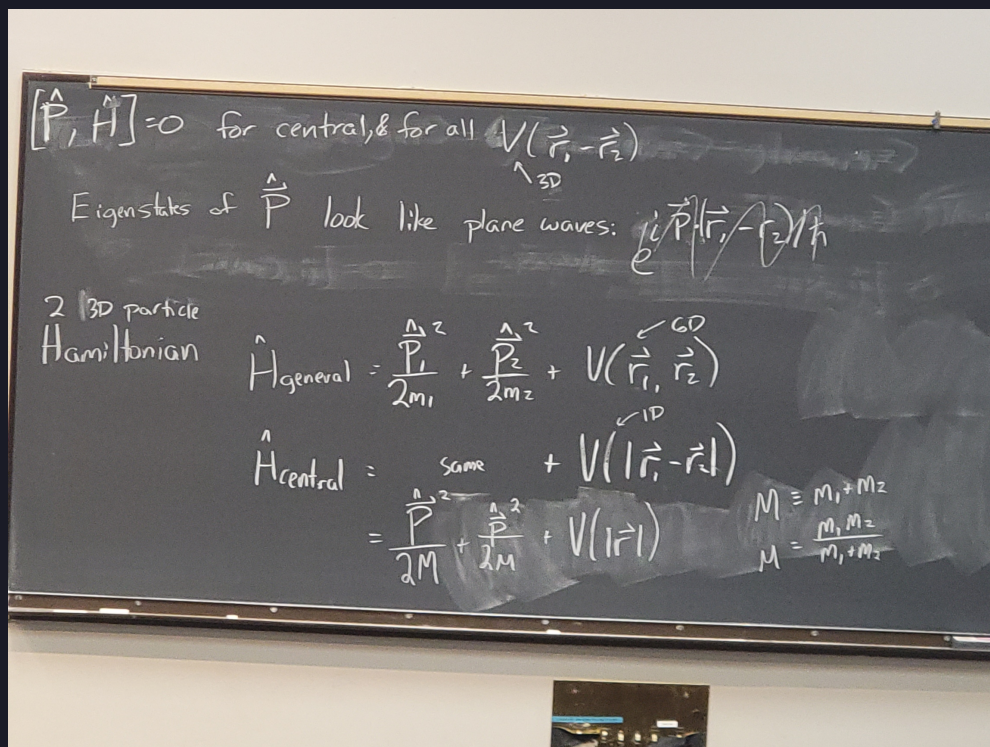
multiparticle commutations

$$[\hat{r}_{ni}, \hat{p}_{nj}] = i\hbar \delta_{ij} \quad (1)$$

(2)

And all actions between different particles commute (what about entanglement?)

We can write out a two particle system with the total position and relative position instead of two total positions.



$\frac{1}{\sqrt{2}}|a, b=+1\rangle + \frac{1}{\sqrt{2}}|a, b=-1\rangle$

$\hat{A}|\psi(t)\rangle = \hat{A}e^{-i\hat{H}t/\hbar}|a\rangle$
 iff $[\hat{A}, \hat{H}] = 0$ then
 $= e^{-i\hat{H}t/\hbar} \hat{A}|a\rangle = a|\psi(t)\rangle$

$[\hat{r}_i, \hat{p}_j] = i\hbar \delta_{ij}$
 $[\hat{r}_1, \hat{p}_1] = i\hbar$
 $[\hat{r}_2, \hat{p}_2] = i\hbar$

$[\hat{r}_1, \hat{p}_2] = 0$
 $[\hat{r}_2, \hat{p}_1] = 0$

$\hat{r} = \hat{r}_1 - \hat{r}_2$
 $\hat{p} = \frac{m_2 \hat{p}_1 - m_1 \hat{p}_2}{m_1 + m_2}$
 $\hat{R} = \frac{m_1 \hat{r}_1 + m_2 \hat{r}_2}{m_1 + m_2}$ (c.m. pos)
 $\hat{P} = \hat{p}_1 + \hat{p}_2$
 $[\hat{R}_i, \hat{P}_j] = i\hbar \delta_{ij}$

If $[\hat{A}, \hat{B}] = 0$ then
 I can pick an orthonormal basis
 of simultaneous eigenstates.
 $\hat{A}|a, b\rangle = a|a, b\rangle$
 $\hat{B}|a, b\rangle = b|a, b\rangle$
 $\frac{1}{\sqrt{2}}|a, b=+1\rangle + \frac{1}{\sqrt{2}}|a, b=-1\rangle$

If $|a\rangle$ is a state of definite $\hat{A}$
 $\hat{A}|a\rangle = a|a\rangle$
 $|\psi(t)\rangle = \hat{U}(t)|a\rangle = e^{-i\hat{H}t/\hbar}|a\rangle$
 $\hat{A}|\psi(t)\rangle = \hat{A}e^{-i\hat{H}t/\hbar}|a\rangle$
 iff $[\hat{A}, \hat{H}] = 0$ then
 $= e^{-i\hat{H}t/\hbar} \hat{A}|a\rangle = a|\psi(t)\rangle$

$[\hat{r}_i, \hat{p}_j] = i\hbar \delta_{ij}$
 $[\hat{r}_1, \hat{p}_1] = i\hbar$
 $[\hat{r}_2, \hat{p}_2] = i\hbar$

$[\hat{r}_1, \hat{p}_2] = 0$
 $[\hat{r}_2, \hat{p}_1] = 0$

$\hat{r} = \hat{r}_1 - \hat{r}_2$
 $\hat{p} = \frac{m_2 \hat{p}_1 - m_1 \hat{p}_2}{m_1 + m_2}$
 $\hat{R} = \frac{m_1 \hat{r}_1 + m_2 \hat{r}_2}{m_1 + m_2}$ (c.m. pos)
 $\hat{P} = \hat{p}_1 + \hat{p}_2$
 $[\hat{R}_i, \hat{P}_j] = i\hbar \delta_{ij}$